

14.2 NMF Topic Modeling

Non-negative factors as an additive topic model.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

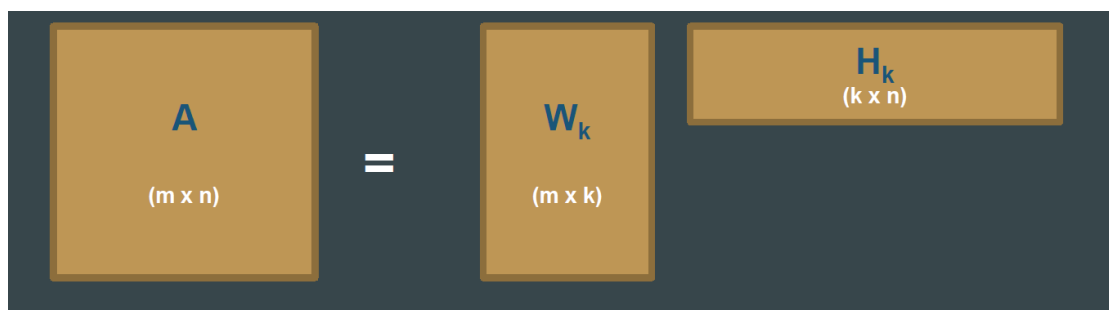
1. Topics that only add

Non-negative matrix factorization (NMF) splits a non-negative matrix ($\mathbf{A}$) (counts or TF-IDF) into two non-negative factors:

$$[\mathbf{A} \approx \mathbf{W}\mathbf{H}.]$$

No minus signs. A document is an **additive** mix of parts. That matches how people talk about topics: this article is some sports plus some politics, not sports *minus* politics.

- ($\mathbf{W}$): **basis**. Each column is a topic as a non-negative combination of terms (*sports, science, politics* as atomic parts).
- ($\mathbf{H}$): **codes**. Each column says how to mix those parts to rebuild one document (30% sports, 50% science, 20% politics).



If ($\mathbf{A}$) is (m) terms by (n) documents and you ask for (k) topics, ($\mathbf{W}$) is ($m \times k$) and ($\mathbf{H}$) is ($k \times n$). Same cartoon as SVD and LDA, different constraints.

2. How it is fit

NMF is an optimization loop, not a generative story.

1. **Initialize** ($\mathbf{W}$) and ($\mathbf{H}$) with non-negative values (random, nndsvd, or a warm start).
2. **Alternate**. Fix ($\mathbf{H}$), update ($\mathbf{W}$) to reduce ($\|\mathbf{A} - \mathbf{WH}\|$) without going negative. Then fix ($\mathbf{W}$) and update ($\mathbf{H}$). Gradient steps or multiplicative updates are the usual engines.
3. **Stop** when the reconstruction error flattens or you hit a max iteration count.
4. **Read** columns of ($\mathbf{W}$) as topics and columns of ($\mathbf{H}$) as document mixes.

Because the objective is not convex in both factors at once, two random starts can give two different topic lists. Run more than once.

3. Why NMF is popular for text

Property	Why it helps
Non-negativity	Top words are "this topic uses these terms," not a signed loading
Parts-based	Topics look like additive ingredients, including subtopics
Often sparse	A document uses a few columns of ($\mathbf{W}$), not all (k)
Broad use	Same idea on images and gene-expression; one mental model

NMF topics are often easier to name than SVD topics and cheaper to train than a careful LDA sweep. It is a fair default when you want a factorization and you do not need a probability model.

4. What goes wrong

Limitation	Practice
Columns of ($\mathbf{W}$) are not orthogonal	Topics overlap. That can be honest (a "finance" and "politics" share <i>tax</i>) or mush
You still choose (k)	Same problem as LDA and truncated SVD
Sensitive to init	Several restarts; keep the run with the best error and the most readable words
Cost	Wide TF-IDF and large (k) are slow
Local minima	Restarts again
No negative association	You will not get "this topic is the opposite of that word"

Do not pick NMF because a slide said "disadvantages of SVD." Pick it when you want additive, non-negative parts. Use SVD when you want a stable orthogonal basis and can live with signs. Use LDA when you want a generative mix and held-out likelihood.

scikit-learn's `NMF` on a TF-IDF matrix is the usual first implementation. Look at `reconstruction_err_` **and** the top terms of each component. If two components share eight of their ten words, raise the stop list or lower (k), do not add epochs.

5. NMF next to SVD and LDA

	SVD / LSA	NMF	LDA
Signs	Can be negative	Non-negative	Probabilities
Story	Linear algebra	Additive parts	Generative mix
Typical input	Centered or TF-IDF	Non-negative TF-IDF / counts	Counts
Stability	High	Init-sensitive	Seed-sensitive

Same picture of "terms $\times$ topics $\times$ documents." Different promises to the reader.

6. Practice

1. After NMF, how do you write a one-line name for topic (j)? Which matrix do you sort?
2. Two runs with (k=15) give different top words. What do you change before you change (k)?
3. A client asks for "topics that are completely distinct." Why might NMF disappoint them, and what would you try instead or in addition?